

Number Systems

Core Concepts:

Everything can be modelled by numbers (and some appropriate conversions)

Bits viewed "through different lenses" / "in different contexts" and represent different things

Base 10: Decimal

↓ 0, 1, 2, 3, 4, 5, 6, 7, 8, 9

Positional Counting system:

Rightmost digit - Least Significant digit (LSD)
Leftmost digit - Most Significant digit (MSD)

10^3	10^2	10^1	10^0
	0	0	1
	0	0	2
	0	0	3
	"	...	...
	"	0	9
	"		
etc. etc.	"	1	0
	"	1	1
	"	1	2
	"	...	...
	"	1	9
	"		
	"	2	0
	"	2	1
	"		
	"	9	9
	"		
	1	0	0
	1	0	7

Number Systems

Base 2: Binary

0, 1

On, off

0v, 3.3v/5v

* Number system suitable for machines *

o Base 2 can be used to directly represent data

o Moreover, since numbers can be used to represent data, the base of the number is irrelevant

Our first level of abstraction is to represent binary value of 1 to the transistor's on state and 0 to the transistor's off state

Number system Conversions

Dec₁₀ → Bin₂

Method:

Continually divide by 2
and read remainders
in reverse

2	25	
2	12	1
2	6	0
2	3	0
2	1	1
	0	1

$$25_{10} = 11001_2$$

Each binary digit is a bit

Bin₂ → Dec₁₀

$$\begin{array}{r} 11001 \\ \begin{array}{l} 2^0 \times 1 \\ 2^1 \times 0 \\ 2^2 \times 0 \\ 2^3 \times 1 \\ 2^4 \times 1 \end{array} \end{array} \begin{array}{r} 1 \\ 0 \\ 0 \\ 8 \\ 16 \\ \hline 25 \end{array}$$

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
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Number Systems

To convert to a base from decimal,
divide by the value of the base